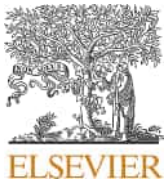
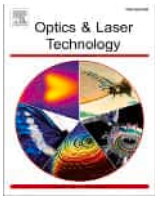
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Full length article

Information-quantitative evaluation of linear computational imaging and application in ghost imaging

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HIGHLIGHTS

- We establish a framework for evaluating and optimizing the intrinsic imaging capabilities of linear computational imaging systems, with no requirements on imaging results or the ground truth.
- The framework enables real-time evaluation no longer dependent on specific post-processing procedures, and also provides feedback during the sampling process, facilitating information-driven adaptive imaging.
- The framework is demonstrated using ghost imaging, and can enhance images up to the Cramér–Rao bound, resulting in superior image quality.

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ABSTRACT

Current evaluation of imaging systems mainly relies on comparing the imaging results to the ground truth. However, the ground truth is usually unavailable in practical scenarios. In this paper, we propose a framework for assessing the capabilities of linear computational imaging processes, where the mapping between object and measurements can be modeled as linear. This framework utilizes Bayesian approach to estimate the amount of acquired information from each sampling, independent of the imaging results and ground truth. We demonstrated our framework based on ghost imaging, a typical linear computational imaging system. This-method facilitates image quality to be improved to the Cramér–Rao bound. Furthermore, an adaptive design of the imaging procedure is also developed and integrated into this dynamic evaluation framework to improve the ability to capture information. The proposed framework exhibits intrinsic generalizability to linear computational imaging systems, and its theoretical structure allows for potential extensions to other computational imaging paradigms.

1. Introduction

Computational imaging (CI) provides a novel imaging paradigm, replacing the traditional "one-to-one" direct imaging approach with an indirect measurement and reconstruction process [37,38,66]. A key feature of CI is the joint design of the front-end optics, detection and post-detection processing [4,38]. Consequently, it is feasible to engineer systems that exceed the physical limitations imposed by the optics [1,13,26,27,50]. For instance, CI has been shown to facilitate various applications in imaging, including the enhancement of spatial resolution [54,59], the expansion of the imaging field of view [68], the acquisition of higher-dimensional information with lower-dimensional detectors [51,57], the attainment of target information with a sampling rate that is lower than the Nyquist limit [23,24,60,62], the realisation of imaging at greater distances with fewer photons [31,32], and the flexible design according to specific tasks [11,42,64].

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Despite the prosperous development of CI, it remains essential to establish quantitative evaluation criteria and design standards for CI systems, in view of the complexity and dynamic nature of the imaging process. Current evaluations of imaging systems primarily rely on comparing imaging results to the ground truth, which are all based on the outputs, with the information acquisition process treated as a black box. Furthermore, the ground truth is often unavailable in many scenarios, such as biological imaging [28], making quantitative evaluation of imaging systems difficult. Acknowledging the gravity of this issue, researchers have sought to develop new evaluation methods for imaging systems [9,17,22]. The evaluation of an imaging system's capability is expected to be conducted from the perspective of information acquisition [63]. With the help of information theory, explorations of various concepts in imaging have emerged [14,30,33], including resolution limit [10,40,55], space-bandwidth product [41,67], and information capacity [21]. However, these methods primarily provide a statistical description of a certain imaging capability. How to quantitatively estimate the information obtained for specific operations remains a challenge.

Through actively implementing multiple encoding mechanisms, CI demonstrates enhanced characteristics in information processing. This advancement enables researchers to construct an evaluation framework using information transmission theory, thereby enabling the quantification of the information transfer capacity of imaging systems, and eliminating the requirement for ground truth reference data from physical scenes. In CI, a comprehensive range of imaging processes can be mathematically represented as linear measurement processes, including ghost imaging (GI) [48,49], snapshot compressive imaging [62], computed tomography[47], etc. Such imaging systems are fundamentally characterised by deterministic linear mapping relationships between target characteristics and the acquired measurement data. These imaging methods have been successfully applied in various scenarios, including multi-waveband imaging [34,43], Fourier imaging [7,61], super-resolution imaging [54], hyperspectral imaging [19,35,39], light field imaging [65], compressed imaging [18,36], and medical imaging [45,58], and so on.

In this study, we develop a Bayesian filtering framework to quantitatively evaluate information acquisition capabilities in CI systems. This approach treats CI as a dynamic parameter estimation problem, achieving continuous state vector estimation in evolving systems through joint modeling of target state evolution and state measurement. The established framework is rigorously validated for CI systems leveraging linear measurement processes, with GI—a representative spatiotemporal encoded CI paradigm—serving as the testbed. Our framework enables real-time quantification of imaging information acquisition through iterative computation of Fisher information matrices during measurement. This information-based method eliminates reliance on reference ground truth data, enabling active uncertainty quantification in imaging processes. Based on the assessment framework, we further propose an imaging quality optimization scheme and adaptive measurement strategies for imaging capability enhancement. Both advancements have been validated through GI, demonstrating significant improvements in imaging efficiency. Significantly, numerical simulations and experiments show the framework's effectiveness in guiding imaging process, and the resulting imaging uncertainty can converge to the Cramér–Rao lower bound (CRLB). While current implementations focus on linear imaging systems, potential extensions to other CI methods can be realized through following strategic approaches: the use of local linearization approximation, enhanced Bayesian filtering framework, and Monte Carlo-based numerical calculation implementations [44,46].

2. Methods

An imaging system can be taken as a mapping from the object space to the image space, as shown in Fig. 1(a), through light field. The CI system is an operator, denoted by $\mathcal{H}$, that maps the object information to the measurements. In CI, a wide range of imaging processes can be elegantly modeled through linear measurement processes [3], as

$$\boldsymbol{z} = \mathcal{H}\boldsymbol{x} + \epsilon, \tag{1}$$

where $\boldsymbol{x}$ is a vector in object space representing object information, $\boldsymbol{z}$ is the vector result of measurement, $\mathcal{H}$ represents the process of linear mapping, which can be expressed in matrix form, and ϵ is the noise introduced during the measurement. Imaging fundamentally involves a mapping from object space to image space, with the target information being reconstructed through the inverse problem [52]. This-process can be regarded as a parameter estimation process, in which the value of the target to be measured is estimated based on the statistical distribution of measurement outcomes

$$\mathcal{L}(\boldsymbol{x}) = p\left(\boldsymbol{z}_{1:N} \mid \boldsymbol{x}\right). \tag{2}$$

Through this approach, the desired parameters can be derived from the analysis of experimental data [46]

$$\hat{\boldsymbol{x}} = \arg\max_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}). \tag{3}$$

In this-process, the measured Fisher information is quantifiable. For multi-parameter estimation, the Fisher information is expressed as

$$\boldsymbol{J}(\boldsymbol{x}) = \mathbb{E}\left[-\frac{\partial^2 \ln \mathcal{L}(\boldsymbol{x})}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}\right], \tag{4}$$

indicating that the estimation accuracy of the unknown parameter $\boldsymbol{x}$ depends on the choice of observables and their corresponding measurements $\boldsymbol{z}$ [16]. The CRLB states that the variance of an unbiased estimator is bounded by the reciprocal of the Fisher information [8]. For multiple parameters, the CRLB generalizes into a matrix form derived from the Fisher information.

To address the dynamic nature of the CI process, we propose employing Bayesian filtering as a processing framework for real-time assessment of information acquisition capabilities. Bayesian filtering is a recursive estimation method, which integrates state-evolution and state-measurement models to sequentially estimate the probability density of a dynamic system's state vector [46]. In CI, the target image can be conceptualized as the state vector within the Bayesian filtering framework [3]. The state evolution and measurement in Bayesian filtering are rather similar to the encoding-measurement process in the CI system. Specifically, (1) in general CI systems, the encoded information varies across measurements, reflecting a form of state evolution; (2) each measurement captures encoded image information, mirroring the state-measurement in Bayesian filtering. The probability density estimation of the state vector, derived from the modeled encoding-measurement process, offers a quantitative assessment of the intrinsic imaging performance in CI systems. Furthermore, the recursive nature of Bayesian filtering facilitates real-time estimation of imaging performance and adaptive optimization through the iterative design of subsequent encoding strategies.

In CI, the sensor discretizes measurements through the sampling process, yielding a discrete representation of the mathematical model. This discrete model can be analyzed using vector-matrix formalism. The scene to be imaged, is denoted as a state vector $\boldsymbol{x}_k$. The evolution of the object's state is described by the following discrete-time model:

$$\boldsymbol{x}_k = f_{k-1}(\boldsymbol{x}_{k-1}, w_{k-1}), \tag{5}$$

where f_{k-1} denotes the state transition from time index $(k-1)$ to k, and w_{k-1} represents the process noise sequence. The state-measurement process is expressed as:

$$\boldsymbol{z}_k = h_k(\boldsymbol{x}_k, v_k), \tag{6}$$

where h_k is the measurement function and v_k is the measurement noise sequence. The measurement process corresponds to the mapping process described in Eq. (6). The measurement results are determined by

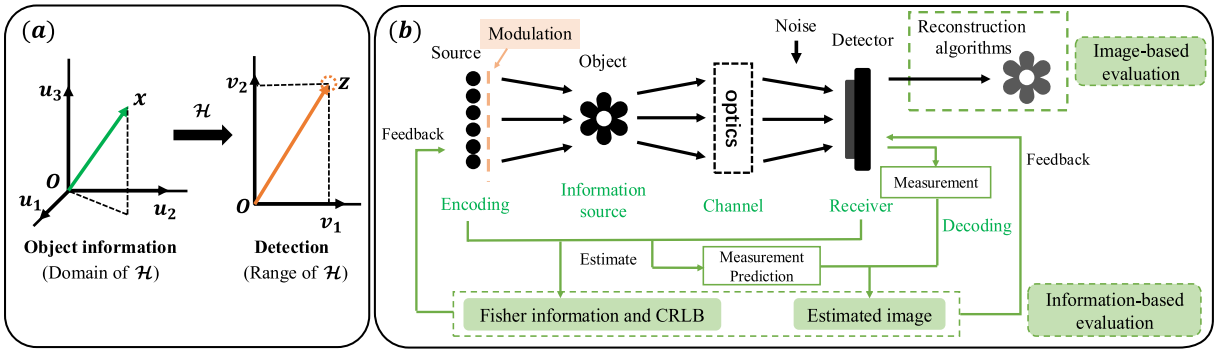


Fig. 1. (a) Imaging can be regarded as a mapping process from one vector space to another vector space [3]. The left graph represents the object space with basis vectors u_n, the right graph represents the measurement space with basis vectors v_n. Note that we have demonstrated a mapping from 3D to 2D space, but the actual dimensionality can be modified. (b) Schematic diagram of a general CI system. The light source is modulated to illuminate the object. The resulting light field, after interacting with the object, propagates through the optical components and is detected by a bucket detector. The object's image is then recovered through the reconstruction algorithms. Conventional image-based evaluation methods rely on the final imaging results for quality assessment. Here, we propose an information-based evaluation method to assess the information obtained in each sampling, independent of both the imaging outcomes and the ground truth. The Fisher information and CRLB are derived from the image model, enabling active quantification of uncertainty in imaging processes. The reconstructed image is estimated by fusing actual measurements with the model-based measurement prediction. Subsequently, the CRLB of measurement errors and the estimated image can be leveraged as feedback to adjust subsequent illumination and detection.

both the physical process of light field propagation and modulation. After the kth measurement, the evaluation incorporates all acquired data $\boldsymbol{Z}_k \triangleq \{\boldsymbol{z}_i, i = 1, \ldots, k\}$, and is iteratively updated with subsequent measurements.

From a Bayesian perspective, the acquisition of an image involves estimating the object's state, while the evaluation of uncertainty in the imaging results quantifies the degree of belief in the state $\boldsymbol{x}_k$. Unlike conventional image evaluation methods, which assess results only after all measurements are completed, our approach enables the evaluation of each encoding and measurement step. This makes it particularly suitable for dynamic encoding in CI. Bayesian filtering employs a two-step recursive process to estimate $p(\boldsymbol{x}_k \mid \boldsymbol{Z}_k)$, the conditional probability density of $\boldsymbol{x}_k$ given $\boldsymbol{Z}_k$. The first step involves predicting the probability density of the state $\boldsymbol{x}_k$ prior to the kth measurement, as described by Eq. (5) (see Appendix A for details)

$$p(\boldsymbol{x}_k \mid \boldsymbol{Z}_{k-1}) = \int p(\boldsymbol{x}_k \mid \boldsymbol{x}_{k-1})\, p(\boldsymbol{x}_{k-1} \mid \boldsymbol{Z}_{k-1})\, \mathrm{d}\boldsymbol{x}_{k-1}. \tag{7}$$

The second step involves updating the prediction based on the measurement $\boldsymbol{z}_k$ via Bayesian estimation (see Appendix A for details)

$$p(\boldsymbol{x}_k \mid \boldsymbol{Z}_k) = \frac{p(\boldsymbol{z}_k \mid \boldsymbol{x}_k)\, p(\boldsymbol{x}_k \mid \boldsymbol{Z}_{k-1})}{p(\boldsymbol{z}_k \mid \boldsymbol{Z}_{k-1})}. \tag{8}$$

This-process enables the recursive estimation of the posterior probability distribution of the object state. For an unbiased estimator $\hat{\boldsymbol{x}}_k$ of the $\boldsymbol{x}_k$, we define $\hat{\boldsymbol{x}}_k \triangleq \mathbb{E}[\boldsymbol{x}_k \mid \boldsymbol{Z}_k]$, where $\mathbb{E}[\cdot]$ represents the expectation value. The covariance matrix of $\hat{\boldsymbol{x}}_k$, denoted as P_k, has a lower bound (CRLB) expressed as [56]

$$P_k \triangleq \mathbb{E}[(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)^T] \geq \boldsymbol{J}_k^{-1} = \mathrm{CRLB}_k, \tag{9}$$

where $\boldsymbol{J}_k$ represents the Fisher information matrix. During the estimation process, $\boldsymbol{J}_k$ can be recursively obtained based on the measurement sequence (see Appendix B for details).

GI is a typical CI method, and its imaging procedure is illustrated in Fig. 1(b). Under specific illumination, light from the object is detected after propagating through certain optics. From the view of information theory, GI involves encoding object information, measuring it, and estimating the object based on a sequence of measurements. The object serves as the information source, the bucket detector functions as the receiver, and the coded illumination and reconstruction correspond to the encoding and decoding processes, respectively. This-provides the flexibility to engineer the physical process of imaging, encoding the object's information into the light field and subsequently decoding it. The Fisher information matrix and CRLB are derived from the image model, enabling active quantification of uncertainty in imaging processes. The reconstructed image is estimated by fusing actual measurements with model-based predictions. Subsequently, the CRLB of measurement errors and the reconstructed image can be leveraged as feedback to adjust subsequent illumination and detection.

In GI, the imaging process can be expressed in a linear form, as shown in Eq. (1). The evolution of states, as described in Eq. (5), can be expressed as

$$\boldsymbol{x}_k = A_{k-1}\boldsymbol{x}_{k-1} + w_{k-1}, \tag{10}$$

where A_{k-1} is a matrix defining the object evolution functions. After illumination, the $(k-1)$th interaction between the modulated light field I_{k-1} and the object results in $\boldsymbol{s}_{k-1} = I_{k-1}\boldsymbol{x}_{k-1}$. The relationship between the results of kth and $(k-1)$th interaction is established as $\boldsymbol{s}_k = B_k\boldsymbol{s}_{k-1} + I_k w_{k-1}$, where $B_k = I_k I_{k-1}^{-1} A_{k-1}$. Subsequently, the detector measures the light field after interacting with the object, yielding the measurement result $\boldsymbol{z}_k$. Therefore, the state-measurement process in GI is a linear process, and can be expressed as

$$\boldsymbol{z}_k = H_k\boldsymbol{s}_k + v_k, \tag{11}$$

where H_k is the measurement matrix of detection process. In GI, H_k represents the signal collection of bucket detection, which is a row vector representing the summation of column vectors $\boldsymbol{s}_k$. In practice, noise typically follows a Gaussian distribution $p(w_{k-1}) \sim \mathbb{N}(0, Q_{k-1})$ and $p(v_k) \sim \mathbb{N}(0, R_k)$. We assume that the noise characteristics remain constant throughout a single experiment. It is assumed that the noise characteristics remain constant throughout a single experiment, so the noise in the state evolution and measurement can be expressed as $p(w_{k-1}) \sim \mathbb{N}(0, Q)$ and $p(v_k) \sim \mathbb{N}(0, R)$, respectively.

The specific structure of this Bayesian GI framework is presented in Algorithm 1. This enables us to obtain the estimation result $\hat{\boldsymbol{x}}_k$ and the covariance matrix P_k in a timely manner. N is the number of measurements. The initial values of $\boldsymbol{x}_0$ and P_0 can be set based on prior experience (see Appendix B for details). The initial state $\boldsymbol{s}_0$ is typically set to the initial state of the object $\boldsymbol{s}_0 = I_0\boldsymbol{x}_0$, with $I_0 = E$ (the identity matrix). For clarity, we present Algorithm 1 as a flowchart in Fig. 2. The process begins by initializing the state x_0, s_0, covariance P_0, and setting

Algorithm 1 Bayesian filtering for GI.

Input: Initial state $\boldsymbol{x}_0$ and s_0, initial covariance P_0, state transition matrix A_k, modulated light field I_k, observation matrix H_k, process noise covariance Q_k, measurement noise covariance R_k
Output: Estimated state $\hat{\boldsymbol{x}}_k$ and estimated covariance P_k
for $k = 1$ to N **do**
  Prediction:
  $\hat{\boldsymbol{x}}'_k = A_{k-1}\hat{\boldsymbol{x}}_{k-1}$
  $\hat{s}'_k = B_k\hat{s}_{k-1}$
  $P'_k = A_{k-1}P_{k-1}A_{k-1}^T + Q_{k-1}$
  Update:
  $K_k = P'_k(H_kI_k)^T(H_kI_kP'_k(H_kI_k)^T + R_k)^{-1}$
  $\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}'_k + K_k(\boldsymbol{z}_k - H_k\hat{\boldsymbol{s}}'_k)$
  $\hat{s}_k = I_k\hat{\boldsymbol{x}}_k$
  $P_k = (I - K_kH_kI_k)P'_k$
  return $\hat{\boldsymbol{x}}_k$ and P_k
end for

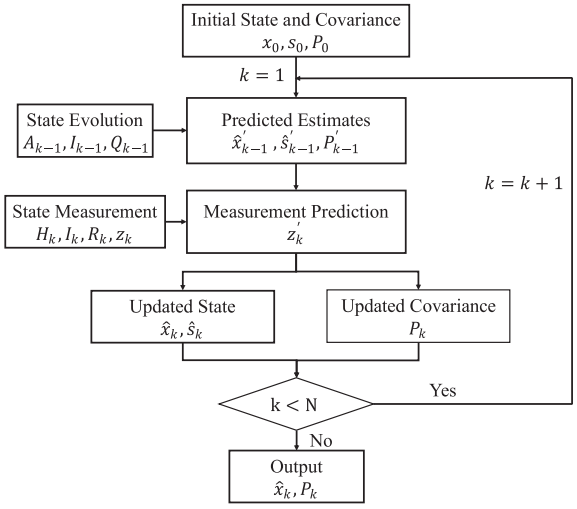


Fig. 2. Flowchart of Bayesian GI framework in Algorithm 1.

the time index to $k = 1$. The algorithm then enters an iterative loop consisting of the following steps: First, the state evolution is used to predict the state estimates $\hat{x}'_{k-1}$, $\hat{s}'_{k-1}$ and the corresponding error covariance $\hat{P}'_{k-1}$. This state evolution model is characterised by matrices A_{k-1}, I_{k-1} and process noise covariance Q_{k-1}. Next, the measurement prediction z'_k is computed using the state measurement model, which is defined by the measurement matrix H_k, the illumination light I_k, and the measurement noise covariance R_k. Subsequently, these predictions are updated using the actual measurement z_k to refine the state $\hat{x}_k$, $\hat{s}_k$ and covariance P_k. Finally, a termination criterion is evaluated: if requirements are met, the loop exits and outputs optimal estimates $\hat{x}_k$ and P_k; otherwise, the time step k is increased, and the loop repeats. This closed-loop iterative process enhances estimation accuracy through successive cycles of prediction and measurement update.

In this-process, the Fisher information matrix is updated iteratively as [44]

$$\boldsymbol{J}_{k+1} = (Q_k + A_k\boldsymbol{J}_kA_k^T)^{-1} + (H_{k+1}I_{k+1})^TR_{k+1}^{-1}H_{k+1}I_{k+1}, \tag{12}$$

which is computed in real-time based on each measurement. The initial Fisher information matrix is given by $\boldsymbol{J}_0 = \mathbb{E}\{P_0^{-1}(\hat{\boldsymbol{x}}_0 - \boldsymbol{x}_0)(\hat{\boldsymbol{x}}_0 - \boldsymbol{x}_0)^T[P_0^{-1}]^T\} = P_0^{-1}$. In linear-Gaussian case, the covariance matrix P_k reaches the CRLB, i.e., [2]

$$\mathrm{CRLB}_k = P_k. \tag{13}$$

This indicates that the estimation process is statistically efficient.

This approach enables the estimation of the image as $\hat{\boldsymbol{x}}_k$ and the evaluation of result uncertainty through CRLB_k. For the kth sampling, the image does not necessarily need to be clear; instead, the amount of achieved information and the associated uncertainty can be estimated with Fisher information matrix and CRLB, respectively. This evaluation is performed concurrently with the sampling process, distinguishing it from existing approaches that typically evaluate results only after all measurements are completed. Moreover, this estimation method is more efficient than traditional algorithms, leading to superior imaging results. By conducting the evaluation during the sampling procedure, it becomes possible to dynamically design subsequent measurements, optimizing the imaging process in real-time.

3. Simulation results

In this-section, we demonstrate our Bayesian filtering framework based on GI [12,13], showing the effectiveness of quantitative characterisation for imaging information acquisition, without requiring physical image results. We also implement the incorporation of prior knowledge into this framework to improve the quality of image reconstruction, and demonstrate adaptive imaging based on real-time information acquisition results.

3.1. Quantitative evaluation of imaging process

In GI, a pseudothermal light I_k with a negative exponential distribution is typically employed for illumination. Since GI collects the object's echo with a bucket detector, the measurement matrix H_k represents the collection of signals. As the object's state is considered to be unchanged, Q is set to zero, and R is the noise level measured in the experiment. For unknown objects, each image pixel is modeled as statistically independent, with an initial value following a Gaussian distribution characterised by expectation x_0 and variance P_0

$$\begin{aligned} &p(\boldsymbol{x}_0) = \mathbb{N}(\boldsymbol{x}_0(i,j), P_0(i,j)),\\ &\boldsymbol{x}_0 = \alpha\boldsymbol{e}, P_0 = \beta E, \end{aligned} \tag{14}$$

where $\boldsymbol{e}$ is unit vector, E is the identity matrix. This means the initial values of each point are the same and without spatial distribution, and the variances of each point are the same and statistically independent. Note that the initial values α and β are specific to each object, initially set to its average reflectance and mean variance. These parameters can be estimated from a limited set of bucket detections. The detailed estimation procedure is provided in Appendix C. Besides, it is assumed that the object's state remains unchanged in the field of view, and A_k can be set as identity matrix $A_k = E$.

In order to demonstrate the efficacy of our approach, we consider the situation at different levels of detected noise. The bucket detection signal-to-noise ratio (BSNR) is defined to characterise the noise level of state measurements in GI [29], as

$$\mathrm{BSNR} = 10\log_{10}\frac{\langle I_b^2\rangle - \langle I_b\rangle^2}{\sigma_n^2}, \tag{15}$$

where I_b represents the bucket detection results and σ_n^2 represents the variance of noise in bucket detection.

In the simulation, the target $\boldsymbol{x}$ is a greyscale image of 64×64 pixels, with a greyscale range from 0 to 10, as shown in Fig. 3 (a). The expected value of the illumination light field I_k is set to 5. In accordance with Algorithm 1, the Bayesian GI results can be obtained under varying BSNR conditions. In GI, for unknown objects, the mean and variance of the image can be estimated from the bucket detection signals. So the initial value of the mean and variance of image can be estimated from limited bucket detection results (see Appendix C for details). Here, the initial value in Eq. (14) is set as $\alpha = 0.181$ and $\beta = 0.134$. The results are illustrated in Fig. 3. The values of R are 131.2, 41.5, 13.1 and 4.1, respectively, representing BSNR as 0 dB, 10 dB, 20 dB and 30 dB. Fig. 3(b)

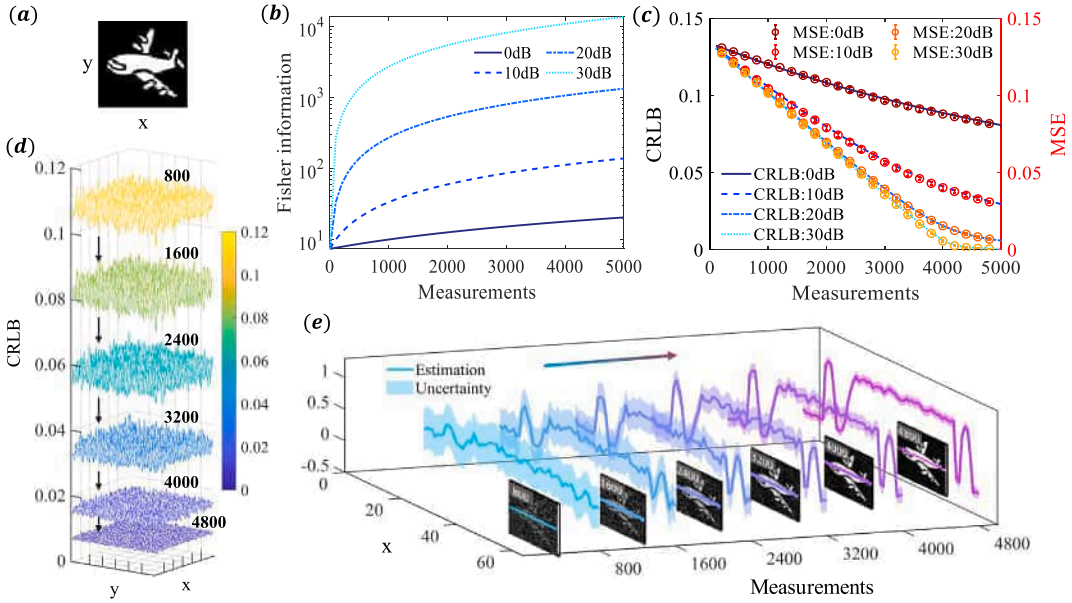


Fig. 3. Imaging results and quantitative evaluation of information acquisition in Bayesian GI. (a) The object being imaged. (b) The average Fisher information of the image varies with the number of measurements, under different BSNRs. (c) The variation of CRLB with the number of measurement on the left axis, and the corresponding MSE of imaging results on the right axis. They are highly compatible. (d) The CRLB of each point varies with different measurements under BSNR is 20 dB. (e) The estimated standard deviation of a one-dimensional cross-section of the corresponding imaging results.

illustrates the variations in the average Fisher information of the image with the number of measurements at different BSNRs. Specifically, in Fisher information and CRLB matrices, the diagonal elements indicate the information content of individual pixels. The results in Fig. 3(b) represent the average value over all pixels.

Meanwhile, the variation in the average CRLB of the image with measurement times under different BSNRs is shown in Fig. 3(c), represented by the left axis. Higher BSNR values lead to a more rapid increase in Fisher information and decrease in the CRLB. To verify our method, we calculated the uncertainty of the specific imaging results, represented by the Mean Square Error (MSE)

$$\mathrm{MSE} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} (\boldsymbol{x}(i,j) - \hat{\boldsymbol{x}}(i,j))^2, \tag{16}$$

where $\boldsymbol{x}$ represents the ground truth, $\hat{\boldsymbol{x}}$ represents the Bayesian GI result, H and W represent the height and width of the image. The vertical axis on the right of Fig. 3(c) denotes the variation in MSE of the image. The data points represent the MSE results at different BSNRs. Each point is obtained through statistics over 10 numerical experiments, with the corresponding error bar provided. As can be seen, the predicted CRLB agrees well with the MSE of the imaging results, which confirms the effectiveness of the CRLB predictions. Furthermore, it also indicates the capability of this-method to reach the theoretical information error bound.

The error for every measured pixel can be estimated based on the specific measurement pattern. Fig. 3(d) shows the CRLB of each point varying with different measurement times, ranging from 800 to 4800, under BSNR of 20 dB. The CRLB of each point gradually decreases and tends to zero as the number of measurements increases, indicating that the estimated imaging result closely approximates the ground truth. Since illumination at each point is random and statistically independent, the decrease in CRLB at each point is essentially identical. To more intuitively represent the estimation error scale of the images, Fig. 3(e) provides the estimated standard deviation of the one-dimensional cross-section of the image. The imaging results in Fig. 3(e) gradually become clearer, and the light areas clearly indicate how the estimated error varies with the number of measurements.

In terms of computational speed, in Fig. 3, the algorithm took 2746 s to estimate the imaging uncertainty for all 5000 measurements. However, the computational frequency could be reduced by calculating the imaging uncertainty every 100 measurements, which would take 28 s (calculated on a Windows 10 computer with an Intel Core i5-12400 processor and 32 GB of memory). This time is within an acceptable range.

3.2. Prior knowledge integration for enhanced image reconstruction

In reality, the structure of the object contains specific characteristics that can be incorporated as a prior in the filtering process. For instance, in most cases, the object is continuous and there exists correlation between adjacent points [25]. Incorporating prior knowledge, we apply some spatial constraints to the estimated image. We assume that each point in the image is correlated with its adjacent points, as illustrated in Fig. 4 (a). The initial covariance matrix in Eq. (14) is then set as $P_0 = \beta E_0 + \gamma\beta E_1 + \gamma\beta E_2 + \gamma\beta E_3 + \gamma\beta E_4$. The items E_1, E_2, E_3, and E_4 in P_0 correspond to the surrounding points of the E_0. The variable γ denotes the correlation between each point and its surrounding points. For simulation, we assign the values of γ to 0 %, 10 %, and 20 %, respectively.

Fig. 4(b) shows the change of CRLB under different correlation constraints γ. As the correlation increases, there is a faster reduction in estimation error. The imaging results under 1000 and 2000 measurements are shown in Fig. 4(c), with the object in Fig. 3(a). It can be seen that as γ increases, the imaging results approach the true value faster. Incorporating prior information accelerates the acquisition of information, and speeds up the image restoration. However, the relationship between γ and image quality is not linear. γ governs the correlation between neighboring pixels. When imaging sparse targets, increasing γ generally improves image recovery. However, if γ is too large, it may result in loss of details or even failure to recover the image. In practical implementation, γ should be set according to experimental conditions and empirical knowledge.

3.3. Dynamic information-driven adaptive imaging

CI allows adjustments to measurements in real time. Here, we present a method for making these adjustments based on the estimated CRLB results. Based on the CRLB, we adopt an adaptive measurement method to enable rapid and efficient image reconstruction.

In natural object images, information is often highly concentrated in the low-frequency range in the frequency domain. We propose to prioritize measuring low frequencies. For instance, the Discrete Cosine Transform (DCT) of the object in Fig. 3(a) is depicted in Fig. 5(a), revealing a concentration of larger values in the lower frequency domain. We propose to encode the object using spatial frequency-limited light. As illustrated by the red dashed box in Fig. 5(a), the spatial frequency of the illuminating light is limited to 25 %, 50 %, and 100 % in the frequency (DCT) domain, respectively. This indicates that the object's information is sampled within a limited frequency range (e.g., 0 %–25 %). Under different frequency encoding ranges, the variations in CRLB are illustrated in Fig. 5(b), incorporating the spatial constraints γ : 20 % in the previous section. As the blue dotted line with encoding frequency range 25 %, the CRLB decreases the fastest initially among these lines, yet asymptotically approaches a constant value, preventing it from reaching zero and consequently, limiting the imaging accuracy. With the encoding frequency range expanded to 50 % and 100 %, the CRLB further decreases, but the initial reduction rate has decelerated. Low-frequency range sampling (25 % and 50 %) is only more efficient in the initial stage. After a certain number of samples, it cannot obtain high-frequency image information, resulting in trending flattening of CRLB.

We designed an adaptive sampling method based on CRLB according to the above characteristics. We propose expanding the encoding frequency range when the CRLB no longer demonstrates further decline. Specifically, an encoding frequency range of 25 % is applied first. When CRLB no longer decreases with an increasing number of samplings, the encoding frequency range is increased to 50 %, and finally, samplings are taken at the full frequency range. The red dotted line in Fig. 5(b) depicts the variations in CRLB with this adaptive imaging strategy. The CRLB maintains a steep decrease initially and ultimately converges to zero.

The corresponding imaging results are presented in Fig. 5(c). The number of measurements is 500–1500 from left to right. The top three rows represent the imaging results with encoding frequency range within 25 %, 50 %, and 100 %, respectively. The narrower the range of frequency domain, the quicker the image emerges, with the most obvious manifestation in the first row of Fig. 5(c). However, information on high frequencies is missing, resulting in noticeable artifacts in the final image. The images in the second and third rows of Fig. 5(c) appear later, but the final resolution is higher. The imaging results of adaptive sampling are presented in the fourth row of Fig. 5(c), exhibiting fast imaging reconstruction at first and high accuracy in the final reconstructed imaging result, eliminating artifacts. This-method enables adaptive design and adjustment of the imaging process independent of imaging results.

4. Experimental results

To further demonstrate the capabilities of our methods, we performed an experimental demonstration using the setup shown in Fig. 6(a). Pseudo-thermal light is generated via a computer-controlled projector (Panasonic: PT-X301), and real-time echo is acquired by a single-pixel detector (Thorlabs: PDA100A2). The reflectance information of the target in the scene is imaged. The distance between the target and the projector is approximately 1.2 m. The projector generates a spatiotemporally varying pseudo-thermal light field that follows a negative exponential distribution and is normalised to 0–1. This is recorded as the illuminance I_k. The object's echo is measured using a single-pixel detector and recorded as the measured value z_k. During the experiment, the intensity of the projector was calibrated, and the noise level R was measured according to different experimental conditions. Through real-time data processing based on Algorithm 1, an image of the target can be obtained, as well as the uncertainty of the estimate.

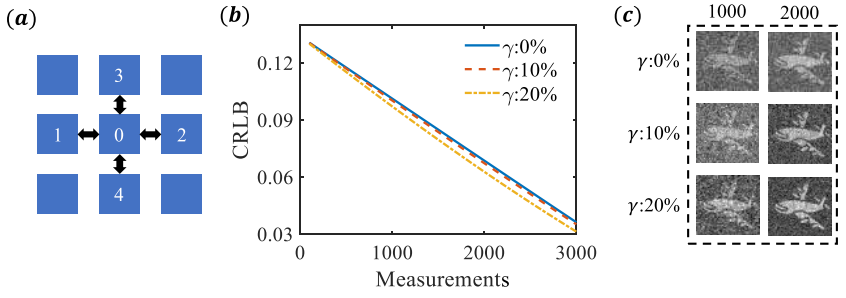


Fig. 4. Bayesian GI results with prior information of spatial constraints. (a) Schematic diagram of the correlation between adjacent pixels. (b) The variation of CRLB with the number of measurements under covariance matrix constraints γ : 0 %, γ : 10 %, and γ : 20 %, respectively. (c) Imaging results under different γ at measurement number of 1000 and 2000.

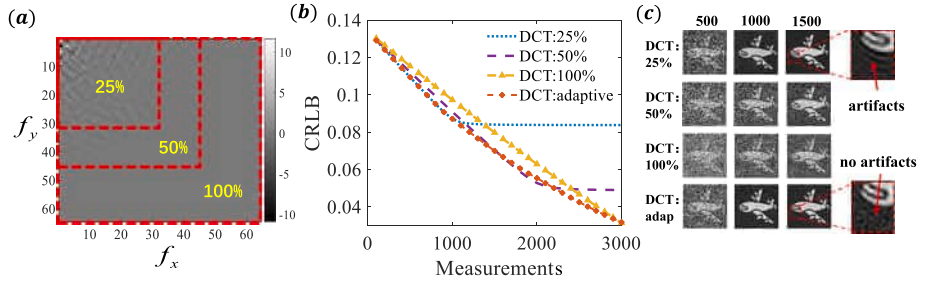


Fig. 5. Adaptive imaging based on the evaluated CRLB. (a) DCT domain representation of the object in Fig. 3(a), the horizontal and vertical axes represent the coordinates of frequency domain. (b) The CRLB varies with the number of measurements under an encoding frequency range of 25 %, 50 %, 100 % in the DCT domain, as well as under adaptive sampling. (c) The imaging results under different encoding frequency range, and adaptive sampling for unknown scenes.

First, we validated the ability to evaluate image quality without actual images. The CRLB is estimated based on the projected speckle field. The object 'A' is a clamping fork (DHC:GCM-5328 M), a component of the experiment. The size of the object is about 7.4 cm × 3.7 cm, and the field of view is 40 × 40 pixels. Object 'B' is a 3D-printed letter 'N'. The dimensions of the latter are approximately 6.0 cm × 4.0 cm, and the field of view is 32 × 32 pixels. As estimated by Algorithm 1, the variations in CRLB during the measurement process of objects 'A' and 'B' are shown in Fig. 6(b) and (c), respectively. In Fig. 6(b), the initial values in Eq. (14) are set as $\alpha = 0.56$ and $\beta = 0.60$. The noise level was measured as $R = 1.5$. In Fig. 6(c), the initial values are set as $\alpha = 1.24$ and $\beta = 1.64$. The noise level was measured as $R = 5.64$. Due to the inherent background illumination of the source (non-zero output at zero input), the reflectivity of the whole scene generates noise. Consequently, the experimentally measured noise level R varies for objects with different reflectance. The results of CRLB are denoted by the solid blue lines, which are represented along the left coordinate axis. To validate the estimated uncertainty, we also calculated the MSE of the image results. In the calculation of MSE, the ground truth is obtained through point scanning with the projector. Notably, the ground truth of the object is usually unavailable in practical scenarios. The results of MSE are denoted by the red asterisks, which are depicted along the right coordinate axis. The variation in MSE results aligns closely with CRLB, thereby validating the efficacy of our method. On an identical computational platform (Windows 10 computer with an Intel Core i5-12400 processor and 32 GB of memory), the algorithm consumed 196 s and 193 s per measurement for two datasets in the experiment, respectively, and calculating imaging uncertainty every 100 measurements would require 2.6 s and 2.5 s, respectively. The algorithm in the experiment consumes significantly less time due to the smaller pixel dimensions of the objects compared to the simulated object in Fig. 3.

In addition, the imaging quality of Bayesian GI is superior to the traditional imaging algorithm. Fig. 7 shows the comparison between the imaging quality of Bayesian GI (BGI), and traditional GI (GI) [6], as

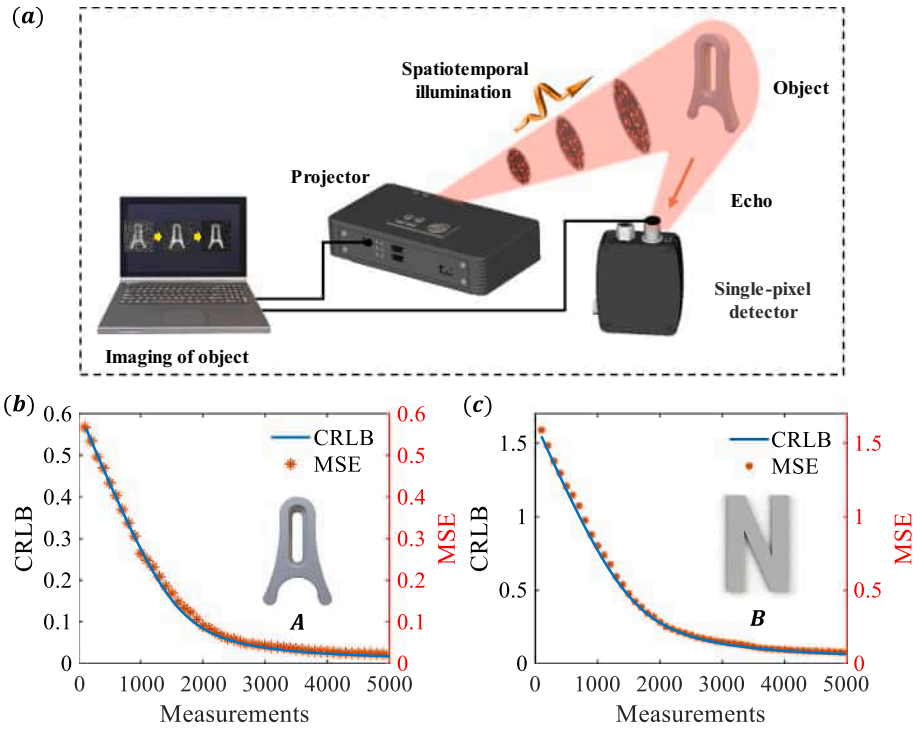


Fig. 6. (a) The experimental setup for Bayesian GI. A projector is employed to project a spatiotemporal modulated pseudothermal speckle light, and the echo of the object is received by a single-pixel detection. (b) and (c) The experimental results of a clamping Fork (denoted as 'A'), and a 3D printed letter (denoted as 'B'), respectively. The predicted CRLB is represented by the solid blue line along the left axis, and the MSE of imaging results is denoted by the red asterisk along the right axis, with respect to the number of measurements.

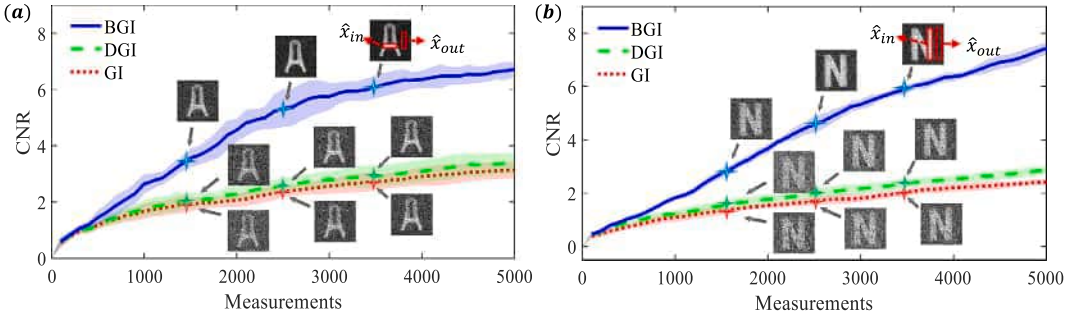


Fig. 7. The experimental BGI results of objects A and B, compared to the imaging results of GI and DGI. (a) and (b) show the experimental CNR results of A and B, respectively. The experimental imaging results of GI, DGI, and BGI at the asterisk position were obtained with the sampling numbers 1500, 2500, and 3500, respectively.

well as differential GI (DGI) [15], which are linear algorithms as well. A quantitative comparison of the imaging quality is shown in Fig. 7, using Contrast-to-Noise Ratio (CNR) as a measure [5]

$$G_{\mathrm{CNR}} = \frac{\langle G(\hat{\boldsymbol{x}}_{in})\rangle - \langle G(\hat{\boldsymbol{x}}_{out})\rangle}{\sqrt{\Delta^2\ G(\hat{\boldsymbol{x}}_{in}) + \Delta^2\ G(\hat{\boldsymbol{x}}_{out})}}, \tag{17}$$

where $\hat{\boldsymbol{x}}_{in}$ represents the area occupied by the object, called object region, and $\hat{\boldsymbol{x}}_{out}$ represents the rest of the area, called background region. G represents the result of the image and $\Delta^2\ G$ is the corresponding variance.

Fig. 7 (a) and (b) represent the results of objects A and B, respectively. The areas within the red box of image in Fig. 7 (a) and (b) are selected as the object regions and background regions. The blue solid line, green dotted line, and red dotted line represent the imaging quality of BGI, DGI, and GI, respectively. Each point is derived from 10 experiments, and the error bars are provided. The imaging quality of BGI exhibits a significant improvement compared to GI and DGI. The imaging results with 1500, 2500, and 3500 samples are depicted in the figure, which is consistent with the trend of the curve. For both object A and object B, the image results of DGI are slightly better than those of GI, and the results of BGI demonstrate a significant improvement over both the GI and DGI results, particularly with a greater number of measurements. These results suggest that BGI demonstrates better image quality.

5. Conclusion and discussion

This paper presents a quantitative evaluation framework for CI systems that can be expressed as a linear process for the first time. This framework can be used to dynamically characterise the specific information acquisition capabilities of these systems. It incorporates Bayesian filtering and employs the Fisher information matrix and the Cramér–Rao lower bound (CRLB) to quantify uncertainty in image retrieval. The proposed framework is applied to GI, a typical linear CI system, through numerical simulations and experiments. The capability and superiority of the proposed framework were validated in three ways: (1) the ability to quantitatively estimate the acquired information and uncertainty (MSE) in images without ground truth; (2) the ability to incorporate prior knowledge to enhance image restoration, and provide real-time feedback during the imaging process to facilitate information-driven adaptive imaging; and (3) the ability to enhance imaging results up to the CRLB, resulting in superior image quality.

Furthermore, by accurately modeling the physical processes of imaging, this framework can introduce as much prior information as possible and achieve optimal estimation. It can be applied to low-light conditions and single-photon imaging [26], both of which require uncertainty to be estimated in weak signals. Through information-driven adaptive imaging, it provides a reliable approach for task-driven imaging [14], thereby achieving higher imaging accuracy. For moving objects, the object's motion information can be modeled in the state-evolution model of Bayesian CI, enabling tracking and imaging within this framework. The presented analysis is primarily applicable to linear CI systems, which are both practical and mathematically straightforward. Other imaging processes can be addressed through linearization techniques, but this requires further analysis [46]. For more complex CI scenarios, integrating Monte Carlo methods into our framework is a feasible solution [44].

CRediT authorship contribution statement

Long-Kun Du: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Chenyu Hu:** Writing – review & editing, Methodology, Data curation. **Zhen-Wu Nie:** Writing – review & editing, Methodology, Formal analysis. **Chen Chang:** Writing – review & editing, Visualization, Data curation. **Shuai Sun:** Writing – review & editing, Validation, Formal analysis. **Shuang Liu:** Writing – review & editing, Visualization, Validation, Formal analysis. **Chenjin Deng:** Writing – review & editing, Formal analysis, Data curation. **Zunwang Bo:** Writing – review & editing, Visualization, Formal analysis. **Wei-Tao Liu:** Writing – review & editing, Visualization, Validation, Formal analysis. **Shensheng Han:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Details for prediction and update in bayesian filtering

Bayesian filtering is a recursive estimation method that provides consecutive probability estimation on the state of a dynamic system based on Bayes' theorem

$$p(\boldsymbol{x}|\boldsymbol{z}) = \frac{p(\boldsymbol{z}|\boldsymbol{x})p(\boldsymbol{x})}{p(\boldsymbol{z})}. \tag{18}$$

where $p(\boldsymbol{x})$ and $p(\boldsymbol{z})$ are the probabilities of observing $\boldsymbol{x}$ and $\boldsymbol{z}$ respectively without any given conditions, which are known as the prior probabilities, and $p(\boldsymbol{z}) \neq 0$. $p(\boldsymbol{x}|\boldsymbol{z})$ is a conditional probability of event $\boldsymbol{x}$ occurring given $\boldsymbol{z}$, which is called the posterior probability.

In Eq. (7), $p(\boldsymbol{x}_k|\boldsymbol{x}_{k-1})$ can be determined by Eq. (5). Eq. (8) is derived from Bayes' theorem, where $p\left(\boldsymbol{z}_k \mid \boldsymbol{Z}_{k-1}\right) = \int p\left(\boldsymbol{z}_k \mid \boldsymbol{x}_k\right) p\left(\boldsymbol{x}_k \mid \boldsymbol{Z}_{k-1}\right) d\boldsymbol{x}_k$. $p(\boldsymbol{z}_k|\boldsymbol{x}_k)$ is determined by Eq. (6). $p(\boldsymbol{x}_0|\boldsymbol{z}_0) \triangleq p(\boldsymbol{x}_0)$, representing prior information about the system state before any observations are made. The recurrence relations in Eq. (7) and Eq. (8) serve as the foundation for the optimal Bayesian solution. The knowledge of the posterior density $p\left(\boldsymbol{x}_k \mid \boldsymbol{Z}_k\right)$ enables the optimal state estimate, and the measurement accuracy of the state estimate.

Appendix B. Fisher information and CRLB in bayesian filtering

For estimation of multiple parameters, the Fisher information matrix is

$$[\boldsymbol{J}(\boldsymbol{x})]_{ij} = \mathbb{E}\left[-\frac{\partial^2 \ln p(\boldsymbol{z};\boldsymbol{x})}{\partial \boldsymbol{x}_i \partial \boldsymbol{x}_j}\right], \tag{19}$$

which means the estimation accuracy of unknown parameter $\boldsymbol{x}$ depends on the choice of observables and their measurement $\boldsymbol{z}$ [16]. CRLB states that the variance of an unbiased estimator cannot be lower than the reciprocal of the Fisher information [8]. The CRLB consequently transforms into a matrix derived from Fisher information for multiple-parameters.

In Bayesian filtering, the information matrix $\boldsymbol{J}_k$ can be calculated recursively [53]

$$\boldsymbol{J}_{k+1} = \boldsymbol{D}_k^{22} - \boldsymbol{D}_k^{21}\left(\boldsymbol{J}_k + \boldsymbol{D}_k^{11}\right)^{-1}\boldsymbol{D}_k^{12} \quad (k > 0) \tag{20}$$

where

$$\begin{aligned}
\boldsymbol{D}_k^{11} &= -\mathbb{E}\left\{\nabla_{\boldsymbol{x}_k}\left[\nabla_{\boldsymbol{x}_k}\log p\left(\boldsymbol{x}_{k+1}\mid\boldsymbol{x}_k\right)\right]^T\right\}\\
\boldsymbol{D}_k^{21} &= -\mathbb{E}\left\{\nabla_{\boldsymbol{x}_k}\left[\nabla_{\boldsymbol{x}_{k+1}}\log p\left(\boldsymbol{x}_{k+1}\mid\boldsymbol{x}_k\right)\right]^T\right\}\\
\boldsymbol{D}_k^{12} &= -\mathbb{E}\left\{\nabla_{\boldsymbol{x}_{k+1}}\left[\nabla_{\boldsymbol{x}_k}\log p\left(\boldsymbol{x}_{k+1}\mid\boldsymbol{x}_k\right)\right]^T\right\} = \left[\boldsymbol{D}_k^{21}\right]^T\\
\boldsymbol{D}_k^{22} &= -\mathbb{E}\left\{\nabla_{\boldsymbol{x}_{k+1}}\left[\nabla_{\boldsymbol{x}_{k+1}}\log p\left(\boldsymbol{x}_{k+1}\mid\boldsymbol{x}_k\right)\right]^T\right\}\\
&\quad -\mathbb{E}\left\{\nabla_{\boldsymbol{x}_{k+1}}\left[\nabla_{\boldsymbol{x}_{k+1}}\log p\left(\boldsymbol{z}_{k+1}\mid\boldsymbol{x}_{k+1}\right)\right]^T\right\}.
\end{aligned} \tag{21}$$

The recursions start with the initial information matrix $\boldsymbol{J}_0$, derived from the initial density $p(\boldsymbol{x}_0)$

$$\boldsymbol{J}_0 = \mathbb{E}\left\{\left[\nabla_{\boldsymbol{x}_0}\log p\left(\boldsymbol{x}_0\right)\right]\left[\nabla_{\boldsymbol{x}_0}\log p\left(\boldsymbol{x}_0\right)\right]^T\right\}. \tag{22}$$

Therefore, we can recursively calculate the Fisher information and CRLB during the Bayesian filtering process.

Appendix C. Setting of initial values in bayesian GI

In Bayesian GI, object information can be estimated from a limited set of bucket detection signals, allowing the initial value $\boldsymbol{x}_0$ and P_0 in Eq. (14) to be determined. The values of α and β can be represented by the average reflectance and variance of a specific target, respectively. The specific method is as follows. The illumination light field is pseudothermal with a negative exponential distribution. Bucket detection can be written as $I_b = \int k_0 I(\rho)\boldsymbol{x}(\rho)d\rho$, where $I(\rho)$ represents the light field that illuminates the surface of object, $\boldsymbol{x}(\rho)$ represents the distribution of object's reflectance, ρ is the position coordinate of the target, and k_0 is a constant. Without loss of generality, k_0 can be considered equal to one and the bucket detection can be written as

$$I_b = \int I(\rho)\boldsymbol{x}(\rho)d\rho, \tag{23}$$

The mean of the object reflectance and α can be expressed as

$$\bar{\boldsymbol{x}} = \frac{\int \boldsymbol{x}(\rho)\mathrm{d}\rho}{A_\mathrm{o}} = \frac{\langle I_b\rangle}{\langle I_o\rangle}\cdot\frac{1}{A_\mathrm{o}}, \quad = \alpha \tag{24}$$

where $\langle I_o\rangle$ represents the mean of the illumination light field, $\langle I_b\rangle$ represents the mean of bucket detection, and A_o is the size of the object area of interest. Besides, the mean square error of the object is

$$\boldsymbol{x}_\mathrm{MSE} = \frac{\int[\boldsymbol{x}(\rho)-\bar{\boldsymbol{x}}]^2\,\mathrm{d}\rho}{A_\mathrm{o}} = \frac{\int \boldsymbol{x}^2(\rho)\mathrm{d}\rho}{A_\mathrm{o}} - \bar{\boldsymbol{x}}^2. \tag{25}$$

According to the circular complex Gaussian characteristics of the pseudothermal light field [20], it can be obtained that

$$\left\langle I_b^2\right\rangle = \langle I_r\rangle^2\left[\int \boldsymbol{x}(\rho)\mathrm{d}\rho\right]^2 + \langle I_o\rangle^2 A_\mathrm{coh}\int \boldsymbol{x}^2(\rho)d\rho, \tag{26}$$

where A_coh represents the coherent area of the illumination light field, i.e., the size of the speckle. Substituting Eqs. (24) and (26) into Eq. (25) yields the variance of object and β

$$\boldsymbol{x}_\mathrm{MSE} = \frac{\left\langle I_b^2\right\rangle - \langle I_b\rangle^2}{\langle I_o\rangle^2 A_\mathrm{o}A_\mathrm{coh}} - \frac{\langle I_b\rangle^2}{\langle I_o\rangle^2}\cdot\frac{1}{A_\mathrm{o}^2}. \quad = \beta \tag{27}$$

Therefore, the initial values α and β in Eq. (14) can be determined.

Appendix D. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.optlastec.2025.113893.

Data availability

Data will be made available upon request.

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